Module 1 — Concepts 1–10

10 foundational ideas. The toolbox of Real Analysis.

1. Neighbourhood of a Point in $\mathbb{R}$

Illustration Standing at 5 on the number line, a neighbourhood is a small bubble around it, say (4.9, 5.1). Everything inside belongs to the neighbourhood.

Definition For $a \in \mathbb{R}$, the **neighbourhood of** a with radius $\varepsilon > 0$ is

$$N_\varepsilon(a) = \{\, x \in \mathbb{R} : |x - a| < \varepsilon \,\}.$$

Plain English All numbers within distance ε of a point.

Problem Find a neighbourhood of 3 with radius 0.2.

2. Bounded Above Set

Illustration In $A = \{1, 2, 3, 4\}$, every element lies below 5. The number 5 acts as a ceiling.

Definition $A \subset \mathbb{R}$ is **bounded above** if $\exists\, M \in \mathbb{R}$ such that

$$x \leq M \quad \text{for all } x \in A.$$

Plain English Some number is bigger than every element of the set.

Problem Is $A = \{n : n \in \mathbb{N}\}$ bounded above?

3. Bounded Below Set

Illustration In $A = \{5, 6, 7, 8\}$, every element is above 4. The number 4 acts as a floor.

Definition A is **bounded below** if $\exists\, m \in \mathbb{R}$ such that

$$m \leq x \quad \forall\, x \in A.$$

Plain English Some number is smaller than or equal to every element of the set.

Problem Is $A = \left\{ \dfrac{1}{n} : n \in \mathbb{N} \right\}$ bounded below?

4. Bounded Set

Illustration The interval $A = [-2, 4]$ is trapped inside a finite box on the number line.

Definition A is **bounded** if $\exists\, M > 0$ such that

$$|x| \leq M \quad \text{for all } x \in A.$$

Equivalently, A is both bounded above and bounded below.

Plain English All elements lie within some finite distance from zero.

Problem Check if $A = \{x : -3 \leq x \leq 10\}$ is bounded.

5. Supremum

Illustration For $A = \{x : x < 5\}$, the number 5 is not in A but is the smallest number greater than everything in A — the tightest ceiling.

Definition $s = \sup A$ if:

(i) $x \leq s$ for all $x \in A$ (upper bound).

(ii) $\forall\, \varepsilon > 0$, $\exists\, a \in A$ with $a > s - \varepsilon$ (no smaller upper bound exists).

Plain English The tightest possible ceiling of a set.

Problem Find $\sup(1, 4)$.

6. Infimum

Illustration For $A = \{x : x > 2\}$, the number 2 is the greatest lower boundary — below every element, yet nothing larger shares this property.

Definition $i = \inf A$ if:

(i) $i \leq x$ for all $x \in A$ (lower bound).

(ii) $\forall\, \varepsilon > 0$, $\exists\, a \in A$ with $a < i + \varepsilon$ (no larger lower bound exists).

Plain English The tightest possible floor of a set.

Problem Find $\inf(2, 6)$.

7. Sequence

Illustration Numbers appearing one after another: $1,\ \frac{1}{2},\ \frac{1}{3},\ \frac{1}{4},\ \ldots$ Each term has a fixed position and value.

Definition A **sequence** is a function $a : \mathbb{N} \to \mathbb{R}$, written $(a_n)_{n=1}^{\infty}$.

Plain English An ordered list of numbers generated by a rule.

Problem Write the first 5 terms of $a_n = \dfrac{1}{n}$.

8. Convergent Sequence

Illustration The sequence $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots$ moves closer and closer to 0.

Definition (a_n) **converges to** L if

$$\forall\, \varepsilon > 0,\ \exists\, N \in \mathbb{N}: \quad n > N \implies |a_n - L| < \varepsilon.$$

Plain English Eventually the terms get arbitrarily close to a specific value L.

Problem Find $\lim_{n\to\infty} \dfrac{1}{n}$.

9. Limit Point of a Set

Illustration In $A = \left\{\frac{1}{n} : n \in \mathbb{N}\right\}$, the terms accumulate near 0. Even though $0 \notin A$, every neighbourhood of 0 contains points of A.

Definition x is a **limit point** of A if

$$\big(N_\varepsilon(x) \setminus \{x\}\big) \cap A \neq \emptyset \quad \forall\, \varepsilon > 0.$$

Plain English Elements of the set cluster infinitely close to the limit point.

Problem Find the limit point(s) of $A = \left\{\dfrac{1}{n} : n \in \mathbb{N}\right\}$.

10. Open Set

Illustration In $(2, 5)$, every point has breathing room — a small neighbourhood around any interior point stays entirely inside the interval.

Definition $U \subseteq \mathbb{R}$ is **open** if

$$\forall\, x \in U,\ \exists\, \varepsilon > 0: \quad (x - \varepsilon,\ x + \varepsilon) \subset U.$$

Plain English No boundary points are included — every point has elbow room.

Problem Is $(1, 4)$ open?

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10 concepts covered. These form the foundation of Real Analysis — nearly everything in the syllabus grows from them.

Module 1 — Concepts 11–20

Sets and sequences under a microscope. Machinery of *Cauchy* and *Weierstrass*.

1. Isolated Point

Illustration In $A = \{1, 2, 3, 7\}$, the point 7 sits alone — zooming near it reveals no other element of A nearby.

Definition $x \in A$ is an **isolated point** of A if $\exists\, \varepsilon > 0$ such that

$$(x - \varepsilon,\ x + \varepsilon) \cap A = \{x\}.$$

Plain English An isolated point has *empty space around it* — no neighbours from the set.

Problem Find the isolated points of $A = \{1, 2, 3\} \cup (5, 6)$.

2. Closed Set

Illustration The interval $[2, 5]$ includes both endpoints. Nothing is "missing" at the edges.

Definition $A \subset \mathbb{R}$ is **closed** if it contains all its limit points. Equivalently, $\mathbb{R} \setminus A$ is open.

Plain English A closed set *contains its edges*.

Problem Is $[1, 4]$ closed?

3. Monotone Sequence

Illustration $1, 2, 3, 4, 5$ always increases. $5, 4, 3, 2, 1$ always decreases. Both move in only one direction.

Definition (a_n) is **monotone increasing** if $a_{n+1} \geq a_n$, and **monotone decreasing** if $a_{n+1} \leq a_n$, for all n.

Plain English A monotone sequence *never reverses direction*.

Problem Is $a_n = \dfrac{1}{n}$ increasing or decreasing?

4. Monotone Convergence Theorem

Illustration The sequence $1,\ 1.5,\ 1.75,\ 1.875, \ldots$ increases steadily but never exceeds 2. It settles near 2.

Definition Every **bounded monotone sequence** in $\mathbb{R}$ is **convergent**.

- Increasing and bounded above $\Rightarrow \lim a_n = \sup\{a_n\}$.
- Decreasing and bounded below $\Rightarrow \lim a_n = \inf\{a_n\}$.

Plain English One direction + bounded limits $\Rightarrow$ must converge.

Problem Does $a_n = 1 - \dfrac{1}{n}$ converge?

5. Subsequences

Illustration From $1, 2, 3, 4, 5, 6, 7$, picking every second term gives the subsequence $1, 3, 5, 7$.

Definition If (n_k) is strictly increasing in $\mathbb{N}$, then (a_{n_k}) is a **subsequence** of (a_n).

Plain English Delete some terms, keep the rest *in order*.

Problem Find a subsequence of $a_n = n$ containing only even numbers.

6. Divergence Criteria

Illustration $1, 2, 3, 4, 5, \ldots$ grows without bound and never settles — it **diverges**.

Definition (a_n) **diverges** if it converges to no finite limit. Key criteria:

(i) Two subsequences converge to *different* limits.

(ii) (a_n) is unbounded.

Plain English The sequence *never settles* near one value.

Problem Does $a_n = n$ converge or diverge?

7. Bolzano–Weierstrass Theorem (Sequences)

Illustration $1, -1, 1, -1, \ldots$ does not converge, yet contains convergent subsequences: $1, 1, 1, \cdots \to 1$ and $-1, -1, -1, \cdots \to -1$.

Definition Every **bounded sequence** in $\mathbb{R}$ has a **convergent subsequence**.

Plain English Even chaotic bounded sequences *hide a convergent pattern* inside.

Problem Show that $(-1)^n$ has a convergent subsequence.

8. Cauchy Sequence

Illustration $1,\ 1.4,\ 1.41,\ 1.414,\ 1.4142, \ldots$ — the terms crowd together progressively.

Definition (a_n) is **Cauchy** if

$$\forall\, \varepsilon > 0,\ \exists\, N: \quad n, m > N \implies |a_n - a_m| < \varepsilon.$$

Plain English Terms eventually become *arbitrarily close to each other.*

Problem Check if $a_n = \dfrac{1}{n}$ is Cauchy.

9. Contractive Sequence

Illustration $1,\ 0.5,\ 0.25,\ 0.125, \ldots$ — each gap is half the previous. The sequence collapses geometrically toward 0.

Definition (a_n) is **contractive** if $\exists\, k \in (0, 1)$ such that

$$|a_{n+1} - a_n| \ \leq\ k\,|a_n - a_{n-1}| \quad \forall\, n \geq 2.$$

Every contractive sequence is Cauchy, hence convergent.

Plain English Successive gaps *shrink by at least a fixed ratio.*

Problem Check if $a_n = \left(\dfrac{1}{2}\right)^n$ is contractive.

10. Completeness Property of $\mathbb{R}$

Illustration $1.4,\ 1.41,\ 1.414,\ 1.4142, \ldots$ approaches $\sqrt{2}$. The limit may be irrational, but it still *exists inside* $\mathbb{R}$.

Definition Every non-empty subset of $\mathbb{R}$ **bounded above** has a **supremum in** $\mathbb{R}$. Equivalently: every Cauchy sequence converges in $\mathbb{R}$.

Plain English $\mathbb{R}$ has *no gaps* — every limit lands somewhere real.

Problem Find $\sup A$ where $A = \{x \in \mathbb{R} : x < 3\}$.

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20 concepts covered. You now control the structure of sets, the behaviour of sequences, and the backbone of rigorous calculus.

Module 1 — Last 8 Topics

The steel beams of Real Analysis. Developed by *Augustin-Louis Cauchy* and *Karl Weierstrass*.

1. Cauchy's Convergence Criterion

Illustration The decimal approximations 1, 1.4, 1.41, 1.414, 1.4142, ... grow closer together. If the terms eventually *stick together tightly*, the sequence converges.

Definition A sequence (a_n) converges if and only if it is **Cauchy** in $\mathbb{R}$:

$$\forall \varepsilon > 0,\ \exists N \in \mathbb{N}: \quad n, m > N \implies |a_n - a_m| < \varepsilon.$$

Plain English In $\mathbb{R}$, a sequence converges precisely when its terms get *arbitrarily close to each other*.

Problem Show that $a_n = \dfrac{1}{n}$ satisfies the Cauchy condition.

2. Contractive Sequence

Illustration The sequence 1, 0.5, 0.25, 0.125, ... rapidly collapses toward 0. Each gap is exactly half the previous one.

Definition A sequence (a_n) is **contractive** if there exists $k \in (0, 1)$ such that

$$|a_{n+1} - a_n| \ \leq\ k\,|a_n - a_{n-1}| \quad \forall\, n \geq 2.$$

Every contractive sequence is Cauchy, hence convergent in $\mathbb{R}$.

Plain English The difference between terms keeps *shrinking by a constant ratio*.

Problem Determine whether $a_n = \left(\dfrac{1}{3}\right)^n$ is contractive.

3. Completeness Property of $\mathbb{R}$

Illustration The numbers 1.4, 1.41, 1.414, 1.4142, ... approach $\sqrt{2}$. Even though no rational equals $\sqrt{2}$, the limit exists *inside* $\mathbb{R}$.

Definition Every non-empty subset of $\mathbb{R}$ that is **bounded above** has a **supremum in** $\mathbb{R}$.
Equivalently: every Cauchy sequence of real numbers converges in $\mathbb{R}$.

Plain English The real number system has *no holes or gaps*.

Problem Find $\sup A$ where $A = \{x \in \mathbb{R} : x < 7\}$.

4. Density of Rational Numbers in $\mathbb{R}$

Illustration Between 2 and 3 lie infinitely many rationals: 2.1, 2.5, 2.75, 2.999, ...

Definition For any $a < b$ in $\mathbb{R}$, there exists a rational $q \in \mathbb{Q}$ such that

$$a < q < b.$$

Plain English Between *any* two real numbers there is always a rational number.

Problem Find a rational number between $\sqrt{2}$ and 2.

5. Density of Irrational Numbers in $\mathbb{R}$

Illustration Between 1 and 2 lie irrational numbers such as $\sqrt{2}$, $\frac{\pi}{2}$, $\sqrt{3}$.

Definition For any $a < b$ in $\mathbb{R}$, there exists an irrational number x such that

$$a < x < b.$$

Plain English No matter how small the gap, *irrational numbers exist inside it.*

Problem Find an irrational number between 1 and 2.

6. Monotone Subsequence Theorem

Illustration The sequence $5, 1, 4, 2, 3$ looks messy, yet contains the increasing subsequence $1, 2, 3$ — hidden order inside apparent chaos.

Definition Every sequence (a_n) in $\mathbb{R}$ has a **monotone subsequence** (either non-decreasing or non-increasing).

Plain English Inside every sequence there is *a subsequence that goes only up or only down.*

Problem Find an increasing subsequence of 3, 1, 4, 2, 5.

7. Bolzano–Weierstrass Theorem (for Sets)

Illustration Scatter infinitely many points inside $[0, 1]$. No matter how you place them, some cluster must form — the points cannot all stay far apart.

Definition Every **infinite bounded subset** of $\mathbb{R}$ has at least one **limit point**.

Plain English Infinitely many points squeezed into a bounded region *must crowd together* somewhere.

Problem Show that $A = \left\{ \frac{1}{n} : n \in \mathbb{N} \right\}$ has a limit point.

8. Cauchy Sequence (Revisited in Context)

Illustration The approximations to π: 3, 3.1, 3.14, 3.141, 3.1415, ... The terms grow *closer and closer together* at every stage.

Definition (a_n) is **Cauchy** if

$$\forall\, \varepsilon > 0,\ \exists\, N: \quad n, m > N \implies |a_n - a_m| < \varepsilon.$$

In $\mathbb{R}$: Cauchy $\iff$ Convergent (Completeness).

Plain English After some stage, *all terms lie extremely close to each other.*

Problem Show that $a_n = \dfrac{1}{n}$ is Cauchy.

Module 1 Complete. All 28 topics covered. This module builds the logical skeleton of calculus — boundedness, convergence, completeness, and density — as formalised by *Cauchy* and *Weierstrass*.

Module 2 — Infinite Series

What happens when we add infinitely many numbers together. Ideas refined by *Cauchy* and *Leibniz*.

1. Infinite Series

Illustration Adding numbers forever: $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots$ Each term shrinks, so the sum slowly approaches 2.

Definition An **infinite series** is $\sum_{n=1}^{\infty} a_n$, defined via **partial sums**

$$S_n = a_1 + a_2 + \cdots + a_n.$$

Plain English Adding infinitely many numbers step by step.

Problem Write the first 4 partial sums of $\sum_{n=1}^{\infty} \frac{1}{n}$.

2. Convergence and Divergence of Series

Illustration $\sum \frac{1}{2^n}$ has partial sums approaching 2 — *convergent*. $\sum \frac{1}{n}$ keeps growing without bound — *divergent*.

Definition $\sum a_n$ **converges** if

$$\lim_{n\to\infty} S_n = S \quad \text{exists and is finite.}$$

Otherwise it **diverges**.

Plain English If the running total settles near a fixed number, the series converges.

Problem Does $\sum_{n=1}^{\infty} \frac{1}{n}$ converge or diverge?

3. Comparison Test

Illustration $\frac{1}{n^2+1}$ is smaller than $\frac{1}{n^2}$, which is known to converge. So the original series must also converge.

Definition Let $0 \le a_n \le b_n$ for all large n.

- $\sum b_n$ converges $\Rightarrow \sum a_n$ converges.
- $\sum a_n$ diverges $\Rightarrow \sum b_n$ diverges.

Plain English Smaller than a convergent series $\Rightarrow$ convergent.

Problem Test $\sum \frac{1}{n^2+1}$.

4. Limit Comparison Test

Illustration When direct comparison is awkward, compare the long-run *ratio* of two series.

Definition If $a_n, b_n > 0$ and

$$\lim_{n\to\infty} \frac{a_n}{b_n} = c, \quad 0 < c < \infty,$$

then $\sum a_n$ and $\sum b_n$ both converge or both diverge.

Plain English Similar long-run behaviour $\Rightarrow$ same convergence outcome.

Problem Test $\sum \frac{n}{n^3+1}$.

5. Ratio Test

Illustration Look at how fast successive terms shrink by computing $\frac{a_{n+1}}{a_n}$. A ratio below 1 means rapid shrinkage — convergence.

Definition Let $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$.

- $L < 1 \quad \Rightarrow \quad$ converges absolutely.
- $L > 1 \quad \Rightarrow \quad$ diverges.
- $L = 1 \quad \Rightarrow \quad$ inconclusive.

Plain English If each term shrinks much faster than the previous, the series converges.

Problem Test $\sum \frac{1}{n!}$.

6. Root Test

Illustration Instead of ratios, examine $\sqrt[n]{a_n}$ to measure the long-term growth rate.

Definition Let $L = \lim_{n\to\infty} \sqrt[n]{|a_n|}$.

- $L < 1 \quad \Rightarrow \quad$ converges absolutely.
- $L > 1 \quad \Rightarrow \quad$ diverges.
- $L = 1 \quad \Rightarrow \quad$ inconclusive.

Plain English Measures how fast terms shrink when raised to the power $\frac{1}{n}$.

Problem Test $\sum \left(\frac{1}{2}\right)^n$.

7. Integral Test

Illustration Think of a series as areas of unit-width rectangles. If the curve underneath has finite area, the series converges.

Definition Let f be continuous, positive, and decreasing on $[1, \infty)$ with $f(n) = a_n$. Then

$$\sum_{n=1}^{\infty} a_n \text{ converges} \iff \int_1^{\infty} f(x)\,dx \text{ converges.}$$

Plain English Compare the series to the area under a related curve.

Problem Test $\sum \frac{1}{n^2}$ using the integral test.

8. Alternating Series

Illustration $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$ — terms bounce between positive and negative, slowly closing in.

Definition An **alternating series** has the form

$$\sum_{n=1}^{\infty} (-1)^{n+1} a_n, \quad a_n > 0.$$

Plain English Numbers add and subtract alternately.

Problem Write the first 5 terms of $\sum_{n=1}^{\infty} (-1)^n \frac{1}{n}$.

9. Leibniz Test (Alternating Series Test)

Illustration $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$ slowly settles because the terms decrease to zero.

Definition If (a_n) is **decreasing** and $a_n \to 0$, then

$$\sum_{n=1}^{\infty} (-1)^n a_n \quad \text{converges.}$$

Plain English Alternating terms shrinking to zero $\Rightarrow$ series converges.

Problem Test convergence of $\sum_{n=1}^{\infty}(-1)^n \frac{1}{n}$.

10. Absolute and Conditional Convergence

Illustration $\sum \frac{(-1)^n}{n}$ converges, but $\sum \frac{1}{n}$ diverges. So the alternating series is only *conditionally* convergent.

Definition

- **Absolutely convergent:** $\sum |a_n|$ converges.
- **Conditionally convergent:** $\sum a_n$ converges but $\sum |a_n|$ diverges.

Plain English Absolute convergence means the series converges even if all signs become positive.

Problem Is $\sum \frac{(-1)^n}{n^2}$ absolutely or conditionally convergent?

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10 concepts covered. These are the core engines of Module 2 — with them you can test almost any infinite series.

Module 2 — Series Manipulation

How series behave when their terms are manipulated. Ideas of *Cauchy* and *Riemann*.

1. Cauchy Condensation Test

Illustration Rather than checking every term of $\sum \frac{1}{n}$, group them in powers of 2:

$$1 + 2\cdot\tfrac{1}{2} + 4\cdot\tfrac{1}{4} + 8\cdot\tfrac{1}{8} + \cdots$$

The grouped series is easier to analyse.

Definition If (a_n) is positive and decreasing, then

$$\sum_{n=1}^{\infty} a_n \text{ converges} \iff \sum_{n=0}^{\infty} 2^n a_{2^n} \text{ converges.}$$

Plain English Compress the series into doubling blocks — convergence of the original and the condensed series are equivalent.

Problem Use the condensation test to determine whether $\sum \dfrac{1}{n}$ converges.

2. Raabe's Test

Illustration When the ratio test yields $L = 1$ (inconclusive), Raabe's test gives a finer check by measuring how fast the ratio approaches 1.

Definition For a series of positive terms, compute

$$R = \lim_{n\to\infty} n\left(\frac{a_n}{a_{n+1}} - 1\right).$$

- $R > 1 \quad \Rightarrow \quad$ converges.
- $R < 1 \quad \Rightarrow \quad$ diverges.
- $R = 1 \quad \Rightarrow \quad$ inconclusive.

Plain English Measures how strongly terms shrink compared to the harmonic series.

Problem Apply Raabe's test to $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$.

3. Cauchy Product of Series

Illustration Multiplying two infinite series is like multiplying two polynomials — the coefficients combine by a sliding overlap (convolution).

Definition Given $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$, their **Cauchy product** is $\sum_{n=0}^{\infty} c_n$ where

$$c_n = \sum_{k=0}^{n} a_k\, b_{n-k}.$$

If both series converge absolutely, the Cauchy product converges to their product.

Plain English Two series are multiplied term by term in overlapping diagonal layers.

Problem Find the first three terms of the Cauchy product of $\sum_{n=0}^{\infty} x^n$ with itself.

4. Rearrangement of Terms

Illustration Taking $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$ and writing its terms in a different order can change the value of the sum — if the series is not absolutely convergent.

Definition A **rearrangement** of $\sum a_n$ is a new series $\sum a_{\sigma(n)}$, where $\sigma : \mathbb{N} \to \mathbb{N}$ is a bijection (permutation of indices).

Plain English The same numbers, written in a different order in the infinite sum.

Problem Write a rearrangement of the finite sum $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4}$.

5. Riemann Rearrangement Theorem

Illustration The series $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$ converges to $\ln 2$. Yet by reordering its terms one can make it sum to π, or -7, or diverge to $+\infty$.

Definition **Riemann Rearrangement Theorem.** If $\sum a_n$ is *conditionally* convergent, then for any $S \in \mathbb{R} \cup \{\pm\infty\}$ there exists a rearrangement $\sum a_{\sigma(n)}$ that converges to S (or diverges to $\pm\infty$).

Plain English For a conditionally convergent series, changing the order of terms can produce *any* sum whatsoever.

Problem Explain why $\sum \frac{1}{n}$ cannot be rearranged to converge.

Module 2 complete. Three big ideas: when series converge, how to test convergence, and what happens when we manipulate series.

Module 3 — Limits & Continuity

The microscope of calculus. Machinery of *Weierstrass*, *Cauchy*, and *Bolzano*.

1. Limit of a Function (ε–δ Definition)

Illustration As x approaches 2, $f(x) = x^2$ approaches 4 — even if x never equals 2 exactly.

Definition $\lim_{x \to a} f(x) = L$ means: for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon.$$

Plain English If x is very close to a, then $f(x)$ is very close to L.

Problem Find $\lim_{x \to 2}(3x + 1)$.

2. Sequential Criterion for Limits

Illustration Instead of checking all paths toward a, we test every sequence approaching a. If all give the same function limit, the limit exists.

Definition $\lim_{x \to a} f(x) = L$ iff for every sequence $(x_n) \to a$ with $x_n \neq a$,

$$f(x_n) \to L.$$

Plain English Every path approaching a must give the same output limit.

Problem Use sequences to verify $\lim_{x \to 0} x^2 = 0$.

3. Divergence Criteria for Functions

Illustration As $x \to 0$, $f(x) = \frac{1}{x}$ shoots to $\pm\infty$. No finite limit exists.

Definition $\lim_{x \to a} f(x)$ **fails to exist** if there are two sequences $x_n \to a$ and $y_n \to a$ (both $\neq a$) with

$$\lim f(x_n) \neq \lim f(y_n).$$

Plain English If function values do not settle to one number, the limit diverges.

Problem Show that $\lim_{x \to 0} \dfrac{1}{x}$ does not exist.

4. Limit Theorems

Illustration If two functions both have limits, their combinations behave predictably: $\lim(x+2) = \lim x + 2$.

Definition If $\lim_{x\to a} f(x) = A$ and $\lim_{x\to a} g(x) = B$, then

$$\lim(f+g) = A+B, \qquad \lim(fg) = AB, \qquad \lim \tfrac{f}{g} = \tfrac{A}{B}\ (B \neq 0).$$

Plain English Limits respect normal arithmetic operations.

Problem Find $\lim\limits_{x\to 3}(x^2+2x)$.

5. Squeeze Theorem

Illustration A function sandwiched tightly between two others that share a limit must share that limit too.

Definition If $g(x) \le f(x) \le h(x)$ near a and $\lim\limits_{x\to a} g(x) = \lim\limits_{x\to a} h(x) = L$, then

$$\lim_{x\to a} f(x) = L.$$

Plain English Two functions squeezing a third toward the same number force it there.

Problem Find $\lim\limits_{x\to 0} x\sin\left(\dfrac{1}{x}\right)$.

6. One-Sided Limits

Illustration For $f(x) = |x|$, approaching 0 from the left or right gives the same value. For a jump-discontinuity the two sides differ.

Definition **Left limit:** $\lim\limits_{x\to a^-} f(x)$ **Right limit:** $\lim\limits_{x\to a^+} f(x)$.
The two-sided limit exists iff both one-sided limits exist and are equal.

Plain English Check each side separately; agreement means the limit exists.

Problem Find the left and right limits of $f(x) = \dfrac{|x|}{x}$ at $x = 0$.

7. Infinite Limits and Limits at Infinity

Illustration As $x \to \infty$, $f(x) = \frac{1}{x}$ approaches 0. As $x \to 0^+$, $f(x) = \frac{1}{x}$ grows without bound.

Definition **Limit at infinity:** $\lim\limits_{x\to\infty} f(x) = L$ means $|f(x) - L| < \varepsilon$ for all $x > M(\varepsilon)$.
Infinite limit: $\lim\limits_{x\to a} f(x) = \infty$ means $f(x) > K$ whenever $0 < |x-a| < \delta(K)$.

Plain English Either the function settles near a number far out, or it blows up near a point.

Problem Find $\lim_{x\to\infty} \frac{1}{x}$.

8. Continuous Function

Illustration $f(x) = x^2$ can be drawn without lifting the pen — its value at every point matches the limit there.

Definition f is **continuous at** a if

$$\lim_{x\to a} f(x) = f(a).$$

Equivalently: f is defined at a, the limit exists, and they agree.

Plain English The function value matches the limit — no jump, no hole.

Problem Check continuity of $f(x) = x^2$ at $x = 2$.

9. Sequential Criterion for Continuity

Illustration Instead of ε–δ, test sequences converging to the point. If outputs always follow, the function is continuous there.

Definition f is **continuous at** a iff for every sequence $x_n \to a$,

$$f(x_n) \to f(a).$$

Plain English Whenever inputs approach a, outputs must approach $f(a)$.

Problem Verify continuity of $f(x) = x$ using the sequential criterion.

10. Algebra of Continuous Functions

Illustration Combining two smooth curves by addition or multiplication yields another smooth curve.

Definition If f and g are continuous at a, then so are:

- $f + g$ and $f - g$.
- fg.
- f/g, provided $g(a) \neq 0$.

Plain English Continuity survives normal arithmetic operations.

Problem Show that $f(x) = x^2 + 3x$ is continuous everywhere.

11. Continuous Functions on an Interval

Illustration f continuous on $[0, 1]$ means the graph flows smoothly across the entire interval with no breaks.

Definition f is **continuous on** $[a, b]$ if it is continuous at every interior point, continuous from the right at a, and continuous from the left at b.

Plain English No breaks anywhere in the interval.

Problem Is $f(x) = x^3$ continuous on $[-1, 1]$?

12. Intermediate Value Theorem

Illustration A function climbing from 2 to 6 must pass through 4. It cannot skip over values.

Definition If f is continuous on $[a, b]$ and k is any value between $f(a)$ and $f(b)$, then there exists $c \in (a, b)$ such that

$$f(c) = k.$$

Plain English A continuous function cannot jump over values.

Problem Show that $x^3 - x - 1 = 0$ has a root in $(1, 2)$.

13. Location of Roots Theorem

Illustration If $f(1) < 0$ and $f(2) > 0$, the function changes sign, so a root lies somewhere in $(1, 2)$.

Definition If f is continuous on $[a, b]$ and $f(a) \cdot f(b) < 0$, then there exists $c \in (a, b)$ with $f(c) = 0$.
(A direct corollary of the Intermediate Value Theorem.)

Plain English A sign change guarantees a zero between the two points.

Problem Does $f(x) = x^2 - 3$ have a root in $[1, 2]$?

14. Preservation of Intervals

Illustration $f(x) = x^2$ on $[0, 2]$ maps the interval to $[0, 4]$ without gaps — a continuous image of an interval is again an interval.

Definition If f is continuous on an interval I, then $f(I)$ is also an interval.

Plain English Continuous functions cannot produce holes in their range.

Problem Find the range of $f(x) = x^2$ on $[0, 2]$.

15. Uniform Continuity

Illustration Ordinary continuity allows a different δ at each point. Uniform continuity requires *one* δ that works for the entire domain.

Definition f is **uniformly continuous** on a set D if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$x, y \in D,\ |x - y| < \delta \implies |f(x) - f(y)| < \varepsilon.$$

Plain English The function is evenly smooth everywhere — no point demands an exceptionally small δ.

Problem Show that $f(x) = x$ is uniformly continuous on $\mathbb{R}$.

Module 3 complete. Three modules done: structure of numbers, infinite sums, and the rigorous foundations of limits and continuity.

Module 3 — Advanced Topics

Laws governing the terrain of continuous functions. Ideas of *Weierstrass*, *Bolzano*, and *Banach*.

1. Extreme Value Theorem

Illustration A continuous curve on $[-1, 2]$ must reach a peak and a valley. For $f(x) = x^2$: minimum 0 at $x = 0$, maximum 4 at $x = 2$.

Definition If f is continuous on a closed interval $[a, b]$, then there exist $c, d \in [a, b]$ such that

$$f(c) \leq f(x) \leq f(d) \quad \text{for all } x \in [a, b].$$

That is, f attains both its minimum $f(c)$ and maximum $f(d)$.

Plain English A continuous function on a closed interval must attain a maximum and a minimum.

Problem Find the maximum and minimum of $f(x) = x^2 - 2x$ on $[0, 3]$.

2. Fixed Point Idea

Illustration The graphs of $y = \cos x$ and $y = x$ intersect at a point where the function returns its input: $f(x) = x$. That intersection is a fixed point.

Definition A point c in the domain of f is a **fixed point** if

$$f(c) = c.$$

By the Intermediate Value Theorem, any continuous $f : [a, b] \to [a, b]$ has at least one fixed point.

Plain English A fixed point is where the function sends a number back to itself.

Problem Check whether $x = 1$ is a fixed point of $f(x) = \dfrac{x+1}{2}$.

3. Continuous Image of a Compact Interval

Illustration Applying $f(x) = x^2$ to $[0, 2]$ produces $[0, 4]$ — still closed and bounded. No gaps, no escape to infinity.

Definition If f is continuous on a compact (closed and bounded) interval $[a, b]$, then $f([a, b])$ is also compact.
Combined with the Extreme Value Theorem: $f([a, b]) = [f(c), f(d)]$ for some $c, d \in [a, b]$.

Plain English Continuous functions map closed bounded intervals to closed bounded intervals.

Problem Find the range of $f(x) = 2x + 1$ on $[1, 4]$.

4. Uniform Continuity on Closed Intervals

Illustration On $[0, 2]$, the curve $f(x) = x^2$ never changes abruptly. One choice of δ can control the oscillation of f everywhere on the interval simultaneously.

Definition **Heine–Cantor Theorem.** Every function continuous on a closed interval $[a, b]$ is **uniformly continuous** on $[a, b]$: for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$x, y \in [a, b],\ |x - y| < \delta \implies |f(x) - f(y)| < \varepsilon.$$

Plain English On a closed interval, continuity automatically becomes uniform — no point demands a specially small δ.

Problem Is $f(x) = x^2$ uniformly continuous on $[0, 1]$?

Module 3 complete. The logical chain of the module: Limits $\to$ Continuity $\to$ Properties on intervals $\to$ Global behaviour. Architecture of rigorous calculus as built by *Cauchy* and *Weierstrass*.

Module 4 — Differentiation

How functions change locally. Results of *Lagrange* and *Cauchy*.

1. Differentiability at a Point and on an Interval

Illustration Place a tangent line on a smooth curve at a point. The slope of that line is the derivative — the instantaneous rate of change.

Definition f is **differentiable at** a if

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

exists. f is differentiable on an interval if the derivative exists at every point of that interval.

Plain English Differentiability means the function has a well-defined slope at that point.

Problem Is $f(x) = |x|$ differentiable at $x = 0$?

2. Carathéodory's Theorem

Illustration Rather than computing a limit directly, rewrite the change in f as a product involving a continuous factor.

Definition f is differentiable at a **if and only if** there exists a function ϕ continuous at a such that

$$f(x) - f(a) = (x - a)\,\phi(x).$$

In that case $\phi(a) = f'(a)$.

Plain English The change in f can be written as: distance $\times$ smooth factor.

Problem Verify Carathéodory's condition for $f(x) = x^2$ at $a = 1$.

3. Algebra of Differentiable Functions

Illustration Two smooth curves combined by addition or multiplication remain smooth.

Definition If f and g are differentiable, then:

$$(f+g)' = f' + g', \qquad (fg)' = f'g + fg', \qquad \left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2} \ (g \neq 0).$$

Plain English Differentiation behaves predictably under addition, multiplication, and division.

Problem Find the derivative of $f(x) = x^2 + 3x$.

4. Chain Rule

Illustration In $\sin(x^2)$, the inner function x^2 sits inside the outer sin. Differentiating requires peeling both layers.

Definition If $y = f(g(x))$ and both f and g are differentiable, then

$$\frac{dy}{dx} = f'(g(x)) \cdot g'(x).$$

Plain English Differentiate the outer function, then multiply by the derivative of the inner function.

Problem Find $\frac{d}{dx}\sin(x^2)$.

5. Derivative of an Inverse Function

Illustration If a function and its inverse swap input and output, their slopes also invert at corresponding points.

Definition If f is differentiable and invertible near a with $f'(a) \neq 0$, then

$$(f^{-1})'(y) = \frac{1}{f'(x)}, \quad \text{where } y = f(x).$$

Plain English The slope of the inverse is the reciprocal of the original slope.

Problem Find $(f^{-1})'$ if $f(x) = x^3$.

6. Relative Extrema

Illustration A peak in a graph is a local maximum; a valley is a local minimum. At either, the tangent is horizontal — slope zero.

Definition c is a **relative maximum** if $f(c) \geq f(x)$ for x near c; a **relative minimum** if $f(c) \leq f(x)$ for x near c.
First derivative test: if $f'(c) = 0$ and f' changes sign at c, then c is a relative extremum.

Plain English Where the slope is zero and changes sign, the function has a peak or valley.

Problem Find the relative extrema of $f(x) = x^2 - 4x + 3$.

7. Interval Extremum Theorem (Extreme Value Theorem)

Illustration A continuous curve on $[-1, 2]$ cannot float off to infinity; it must reach a definite highest and lowest point somewhere on the interval.

Definition If f is continuous on $[a, b]$, then there exist $c, d \in [a, b]$ such that

$$f(c) \leq f(x) \leq f(d) \quad \text{for all } x \in [a, b].$$

Plain English On a closed interval, a continuous function must attain both its maximum and minimum.

Problem Find the maximum and minimum of $f(x) = x^2$ on $[-1, 2]$.

Topics 1–7 covered. Next: Rolle's Theorem, Mean Value Theorem, Cauchy Mean Value Theorem, and L'Hôpital's Rule — the central engine of calculus proofs.

Module 5 — Taylor's Theorem & Power Series

Polynomials impersonating complicated functions. Ideas of *Taylor*, *Maclaurin*, *Cauchy*, and *Hadamard*.

1. Taylor's Theorem (Lagrange and Cauchy Remainder)

Illustration To approximate a complicated curve near a, build a polynomial that matches the function's value and derivatives at a. More terms $\Rightarrow$ closer approximation.

Definition If f has $n+1$ derivatives near a, then

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + R_n,$$

where the remainder takes two forms:

$$R_n = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1} \quad \text{(Lagrange)}, \qquad R_n = \frac{f^{(n+1)}(c)}{n!}(x-c)^n(x-a) \quad \text{(Cauchy)},$$

for some c between a and x.

Plain English A smooth function equals a polynomial plus a small, controllable error term.

Problem Find the second-degree Taylor polynomial of $f(x) = e^x$ around $x = 0$.

2. Application to Convex Functions

Illustration A convex function (cup opening upward) always lies *above* its tangent line at every point.

Definition f is **convex** on an interval if $f''(x) \geq 0$ there. By Taylor's theorem with Lagrange remainder:

$$f(x) \geq f(a) + f'(a)(x-a) \quad \text{for all } x.$$

Plain English For convex functions the tangent line is always a lower bound for the graph.

Problem Show that $e^x \geq 1 + x$ for all $x \in \mathbb{R}$.

3. Maclaurin Series of the Exponential Function

Illustration Setting $a = 0$ in Taylor's theorem gives a **Maclaurin series** — an infinite polynomial centered at the origin.

Definition

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots, \quad x \in \mathbb{R}.$$

The series converges for all real x.

Plain English The exponential function can be built entirely from powers of x — the infinite polynomial equals the function everywhere.

Problem Use the first three terms of the series to approximate $e^{0.1}$.

4. Maclaurin Series of Trigonometric and Other Functions

Illustration Even oscillating functions like $\sin x$ and $\cos x$ hide infinite polynomials inside them.

Definition

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!},$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!},$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots, \quad |x| \le 1,\ x \ne -1.$$

Plain English Sine, cosine, and logarithm can each be represented as infinite power series, allowing precise numerical computation.

Problem Use the first two non-zero terms of the $\sin x$ series to approximate $\sin(0.2)$.

5. Radius of Convergence

Illustration A power series works perfectly within a certain distance from its center and fails beyond it — like a light bulb illuminating only within a fixed radius.

Definition For $\sum_{n=0}^{\infty} a_n (x-c)^n$, there exists $R \ge 0$ (possibly ∞) such that:

- Series **converges absolutely** if $|x - c| < R$.
- Series **diverges** if $|x - c| > R$.
- Behaviour at $|x - c| = R$ must be checked separately.

R is the **radius of convergence**.

Plain English A power series only converges within a certain distance from its center.

Problem Find the radius of convergence of $\sum_{n=0}^{\infty} x^n$.

6. Cauchy–Hadamard Theorem

Illustration Rather than guessing where a series converges, this theorem gives an exact formula for the radius of convergence.

Definition For $\sum_{n=0}^{\infty} a_n x^n$, the radius of convergence is

$$R = \frac{1}{\limsup_{n\to\infty} |a_n|^{1/n}}.$$

(Set $R = \infty$ if the lim sup is 0, and $R = 0$ if it is ∞.)

Plain English The theorem gives the exact boundary where the series stops converging.

Problem Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n}$ using the Cauchy–Hadamard theorem.

Topic	Core idea
Taylor's theorem	Approximate functions using polynomials
Lagrange remainder	Measures approximation error
Convex functions	Graph lies above the tangent line
Maclaurin series	Taylor series centred at 0
Radius of convergence	Range where the series converges
Cauchy–Hadamard	Formula for the radius

Module 5 complete. Polynomials can impersonate any sufficiently smooth function to arbitrary precision — the central alchemy of analysis.